Laurent Ortiz

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Module 4 Research Analytical Attacks on AES

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Why are related-key attacks considered impractical against full AES in real world situations, but are effective against reduced-round AES 256?

Related-Key attacks are effective against reduced-round AES-256 due to vulnerabilities in it's key schedule that enable differential trails to propagate in early rounds with reasonable complexity; however, these attacks fail to present a threat to the full 14-round AES-256 because of the exponential increase in complexity and unrealistic related-key assumptions. In AES-256, the key expansion algorithm incorporates nonlinear transformations less frequently than in AES-128, leading to prolonged linear relationships between successive round subkeys. This slow diffusion allows attackers to develop differential trails that connect key differences with internal state differences over multiple rounds.

Biryukov explains that by carefully selecting XOR related keys, these structured differences can mitigate internal state differences over eight to ten rounds. It provided effective key recovery attacks with complexities as little as 2^{39} for 9-round versions and 2^{45} for 10-round variants. Previous research analysis of the complete encryption revealed a consistent fundamental issue: the weak diffusion in the AES-256 key schedule. Which allowed for extensive related key differential trails across multiple rounds. The results show that the reduced-round AES-256 is vulnerable not because of brute-force restrictions but because its key schedule design for related key discrepancies can withstand long enough to provide effective differential attacks prior to achieving the complete 14-round diffusion barrier (Biryukov, 2009).

Although related key attacks exploited the structural properties of the AES-256 key schedule, they weren't successful with the attacks on the full 14-round cipher due to the complexity exceeding beyond practical limits and unrealistic assumptions. In a real life scenario, the full related key attack on AES-256 is mathematically impossible because it would take about 2^{131} time and 2^{120} data, with only succeeding for a small fraction of the keys. With the additional rounds beyond simplified versions, and the overall diffusion of the round function prevented uniformed state cancellations from the structured key variations. And because exploiting those vulnerabilities required excessive processing power and attack conditions that are not possible in real systems, the full 14-round AES-256 assured realistic security even while the reduced round variants exposed a flaw in the key scheduling.

Overall, these results show that related key cryptanalysis is more important than just being able to decipher the full AES-256 in practice. The reduced round attacks revealed that security can't be guaranteed by key length alone; the key schedule's design and diffusion properties are also essential. Modern cipher designs have been influenced by the limitations of AES-256's key expansion, specifically its slower diffusion and longer linear relations across the subkeys. Which led to the development of more resilient key schedules that provide faster diffusion and fewer predictable linear dependencies.

References

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